

PAPER_1663 — Jamming Density $\phi_J = \phi_J = 2/(D_{\text{phys}}-1)$ = 2/3 = 0.667

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Granular Matter **Date:** June 18, 2026 **Status:** CLOSED — 0.4% closure (PAPER_134x broader corpus)

Observation

PAPER_1355 (PAPER_134x condensed matter / quantum sector) gives:

$$\phi_J = 2/(D_{\text{phys}}-1) = 2/3 = 0.667$$

Status: **0.4%**.

UQFF Closed Identity

$$\phi_J = 2/(D_{\text{phys}}-1) = 2/3 = 0.667 \quad 0.4\%$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Experimental granular matter measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1355
- Related: PAPER_1646-1655 (tier-23 broader corpus); PAPER_1636-1645 (tier-22 SM)
- Calculator dispatch: `calculate_paradox({"paradox": "jamming_phi_j_2_3"})`

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